

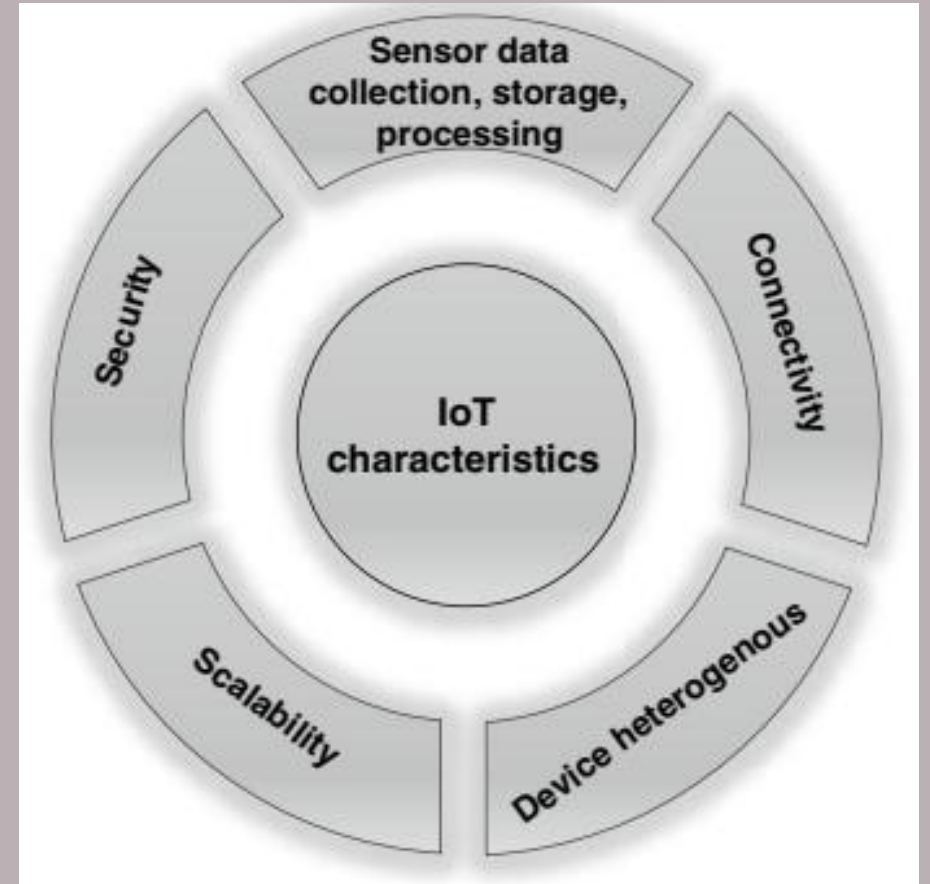
Internet of Things

Fundamental IoT Characteristics

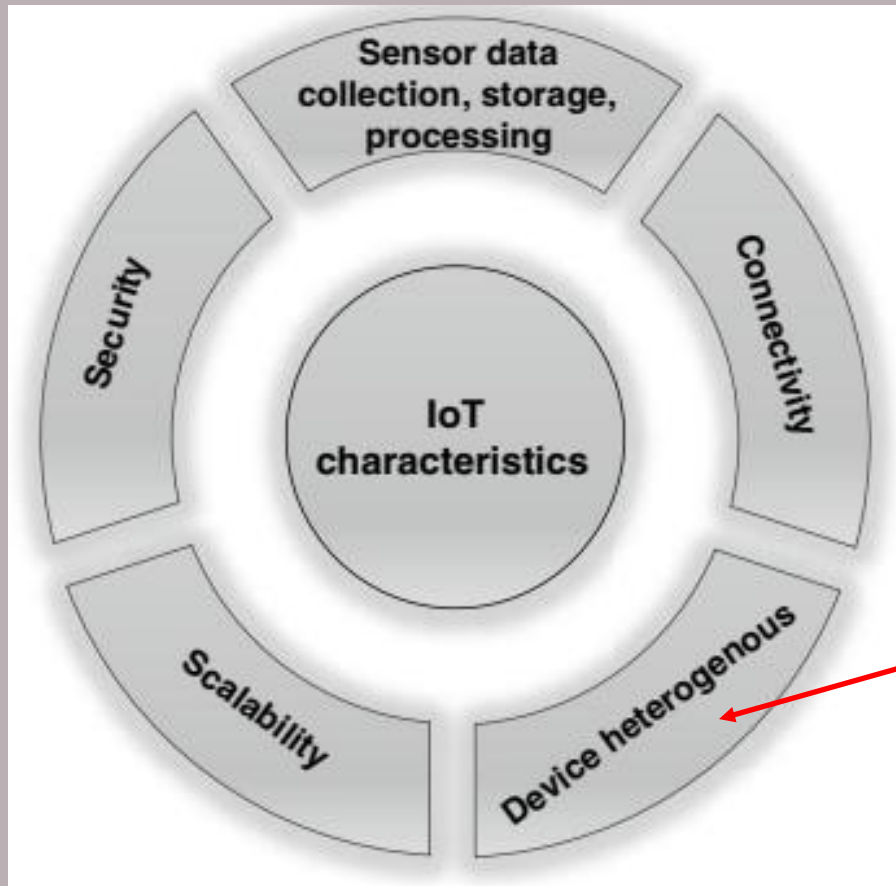
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Fundamental IoT Characteristics

The figure summaries some generic characteristics of the Internet of Things.



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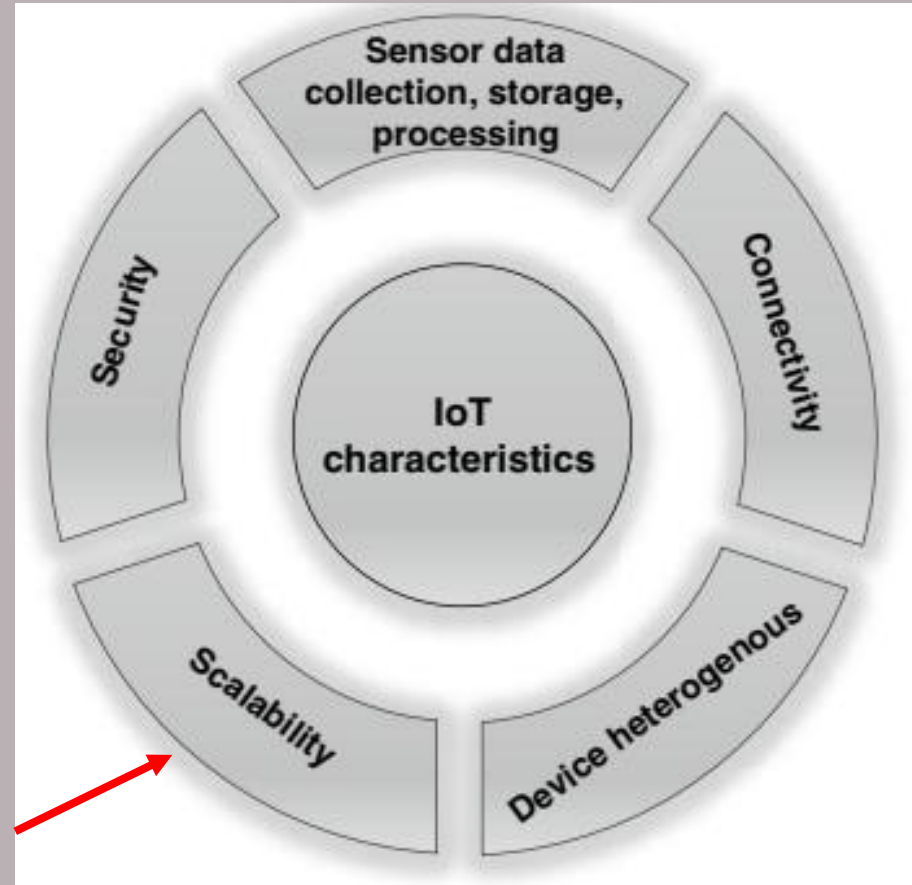
Summary: The key system-level features of the Internet of Things are:

- **Devices heterogeneity:** IoT will be characterized by a large number of heterogeneous devices taking part in the system, which are expected to present very different capabilities from the computational and communication standpoints. The management of such a high level of heterogeneity shall be supported at both architectural and protocol levels.

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Summary: The key system-level features of the Internet of Things are:

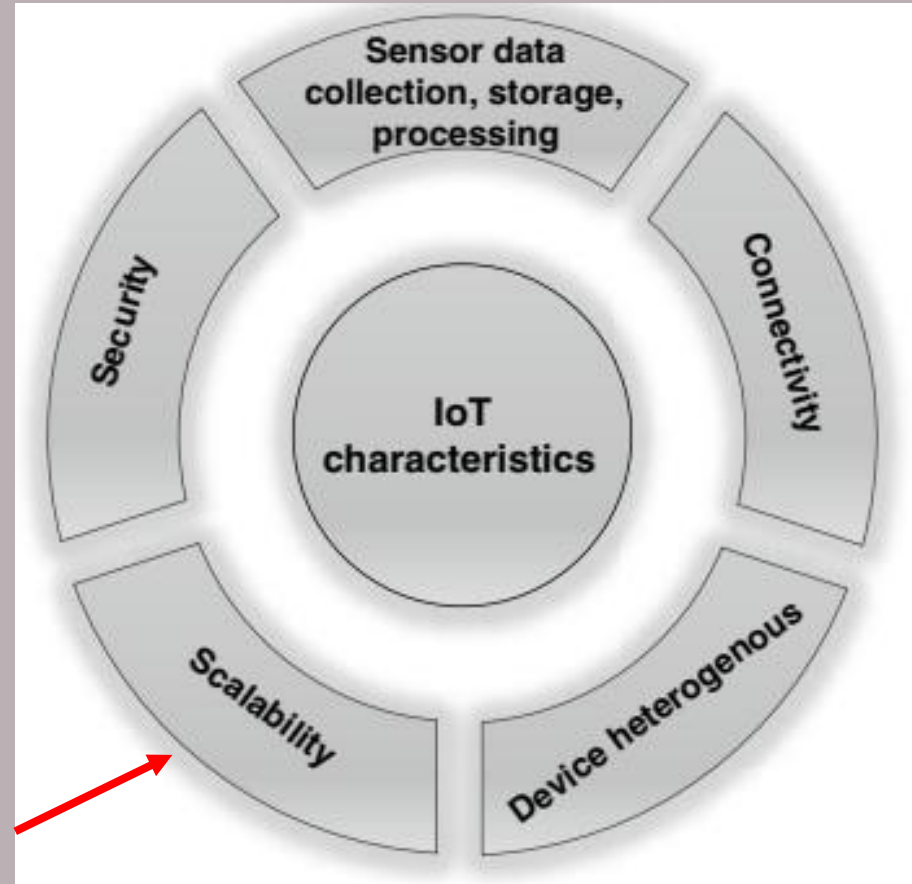
- **Scalability:** As everyday objects get connected to a global information infrastructure, scalability issues arise at different levels, including:
 - i. naming and addressing – due to the sheer size of the resulting system,
 - ii. Data communication and networking – due to the high level of interconnection among a large number of entities,



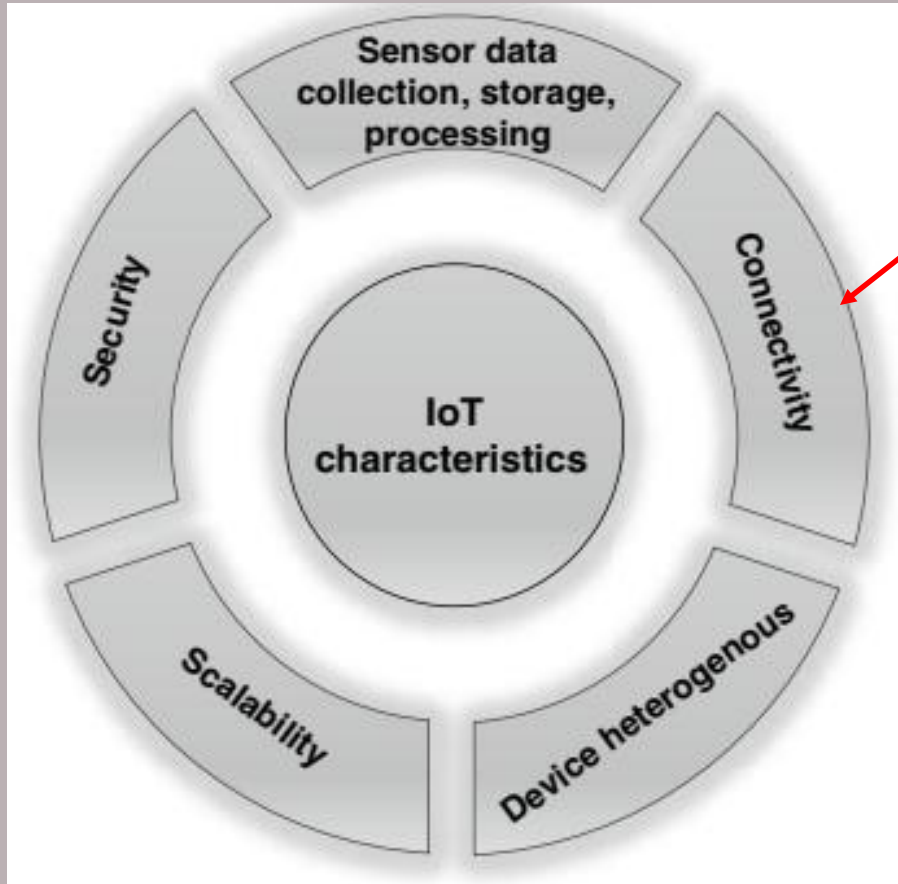
Internet of Things

Summary: The key system-level features of the Internet of Things are:

- **Scalability:** As everyday objects get connected to a global information infrastructure, scalability issues arise at different levels, including:
 - iii. information and knowledge management – due to the possibility of building a digital counterpart to any entity and/or phenomena in the physical realm, and
 - iv. service provisioning and management – due to the massive number of services/service execution options that could be available and the need to handle heterogeneous resources.



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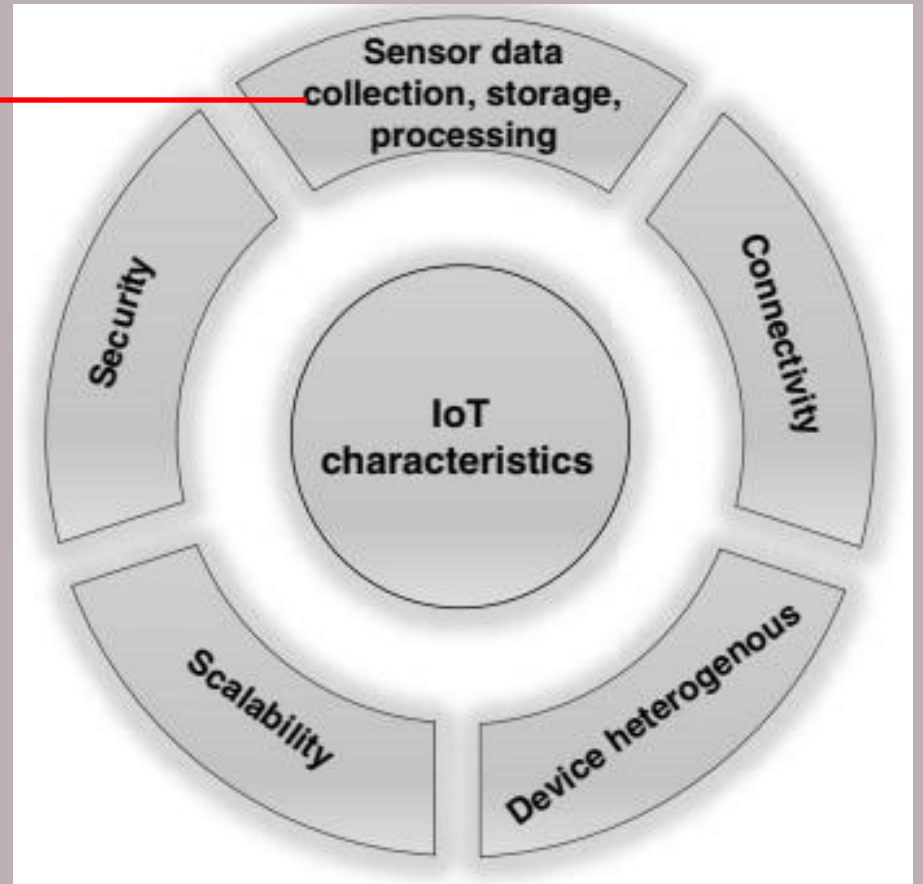
Summary: The key system-level features of the Internet of Things are:

- **Ubiquitous data exchange through proximity wireless technologies:** In IoT, a prominent role will be played by wireless communications technologies, which will enable smart objects to become networked. The ubiquitous adoption of the wireless medium for exchanging data may pose issues in terms of spectrum availability, pushing towards the adoption of cognitive/dynamic radio systems

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Summary: The key system-level features of the Internet of Things are:

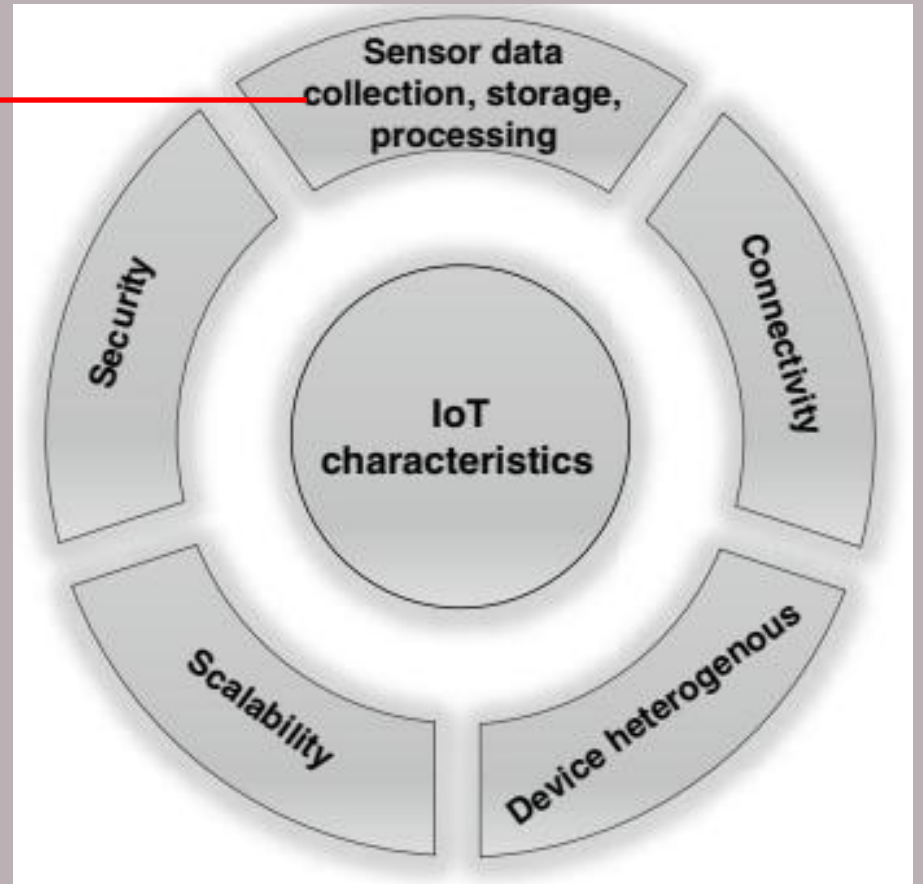
- **Self-organization capabilities:** The complexity and dynamics that many IoT scenarios will likely present calls for distributing intelligence in the system, making smart objects (or a subset thereof) able to autonomously react to a wide range of different situations, in order to minimize human intervention.



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Summary: The key system-level features of the Internet of Things are:

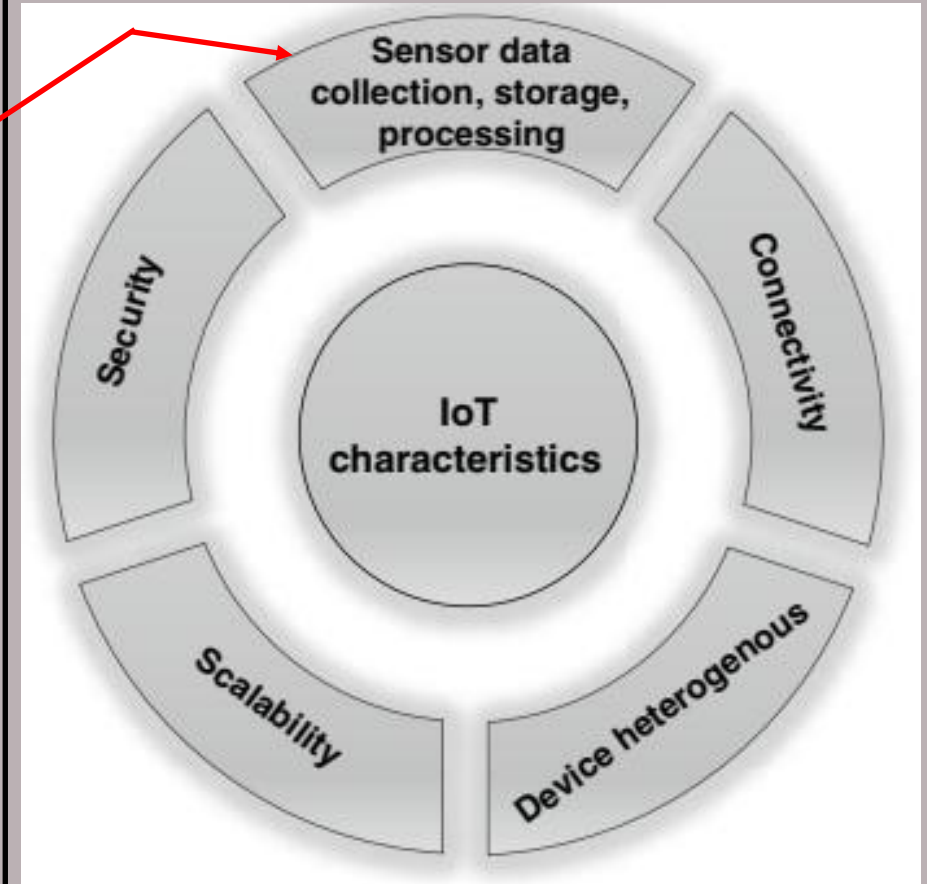
- **Self-organization capabilities:** Following users' requests, nodes in IoT will organize themselves autonomously into transient ad hoc networks, providing the basic means for sharing data and for performing coordinated tasks. This includes ability to perform device and service discovery without requiring an external trigger, to build overlays and to adaptively tune protocols' behavior to adapt to the current context.



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Summary: The key system-level features of the Internet of Things are:

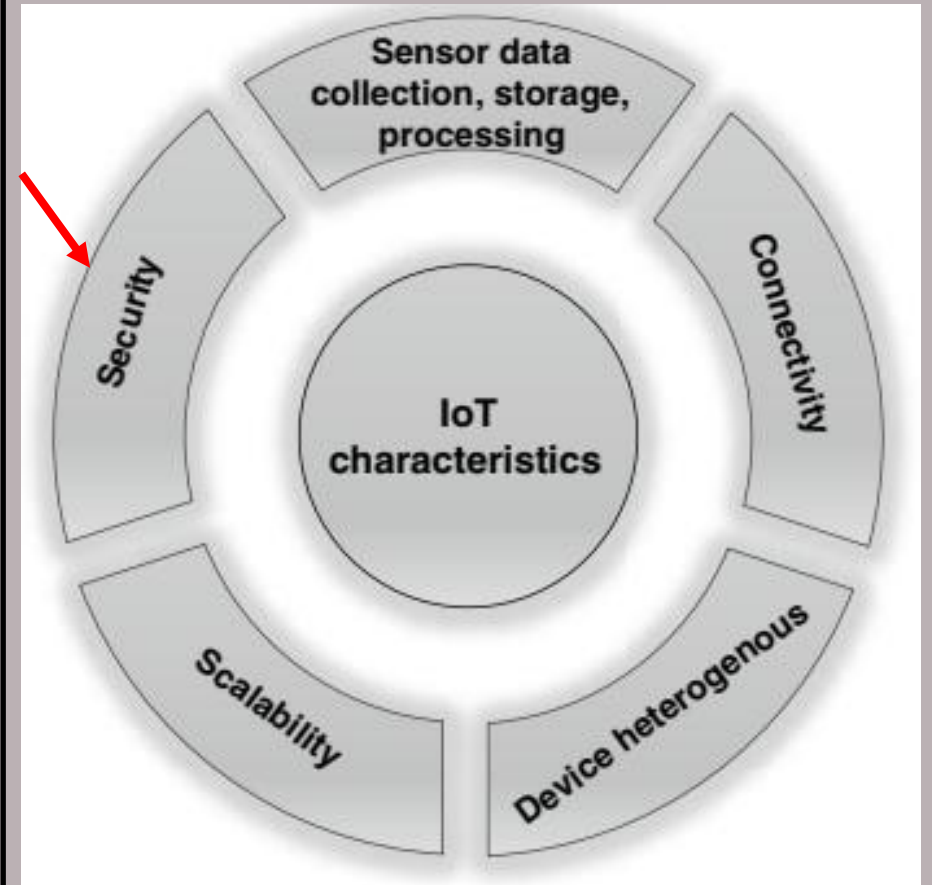
- **Semantic interoperability and data management:** IoT will be much about exchanging and analyzing massive amounts of data. In order to turn them into useful information and to ensure interoperability among different applications, it is necessary to provide data with adequate and standardized formats, models and semantic description of their content (meta-data), using well defined languages and formats. This will enable IoT applications to support automated reasoning, a key feature for enabling the successful adoption of such a technology on a wide scale.



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Summary: The key system-level features of the Internet of Things are:

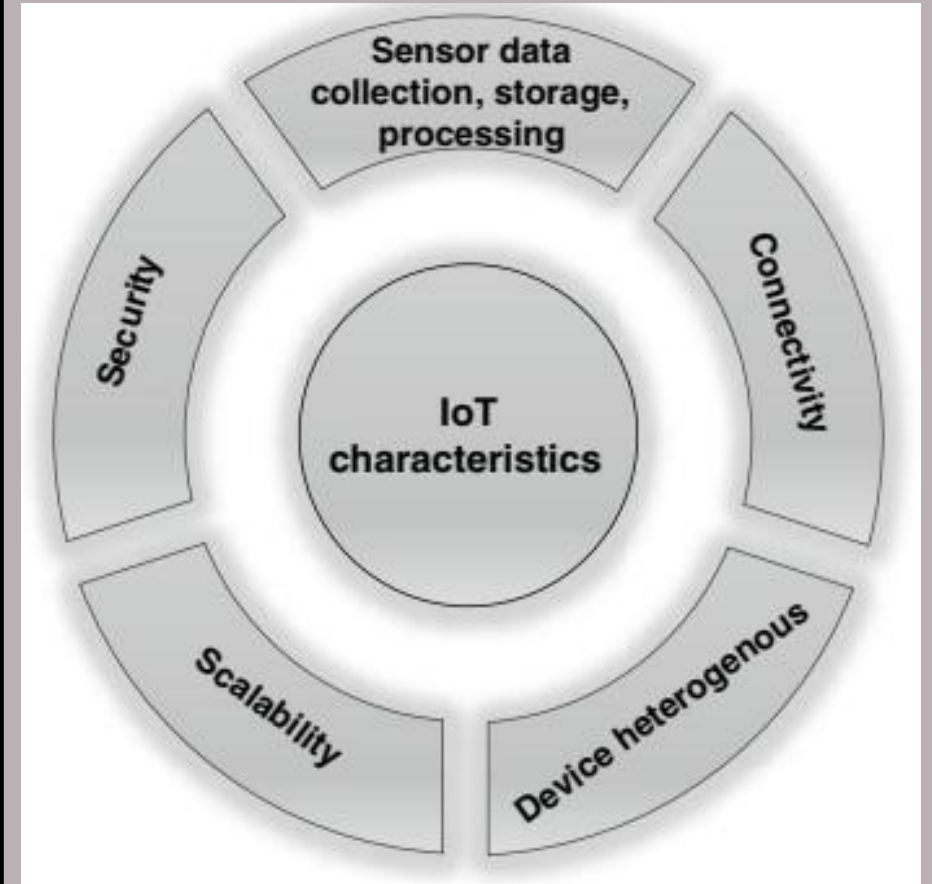
- **Embedded security and privacy-preserving mechanisms:** Due to the tight entanglement with the physical realm, IoT technology should be secure and privacy-preserving by design. This means that security should be considered a key system-level property, and be taken into account in the design of architectures and methods for IoT solutions. This is expected to represent a key requirements for ensuring acceptance by users and the wide adoption of the technology.



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Summary: The key system-level features of the Internet of Things are:

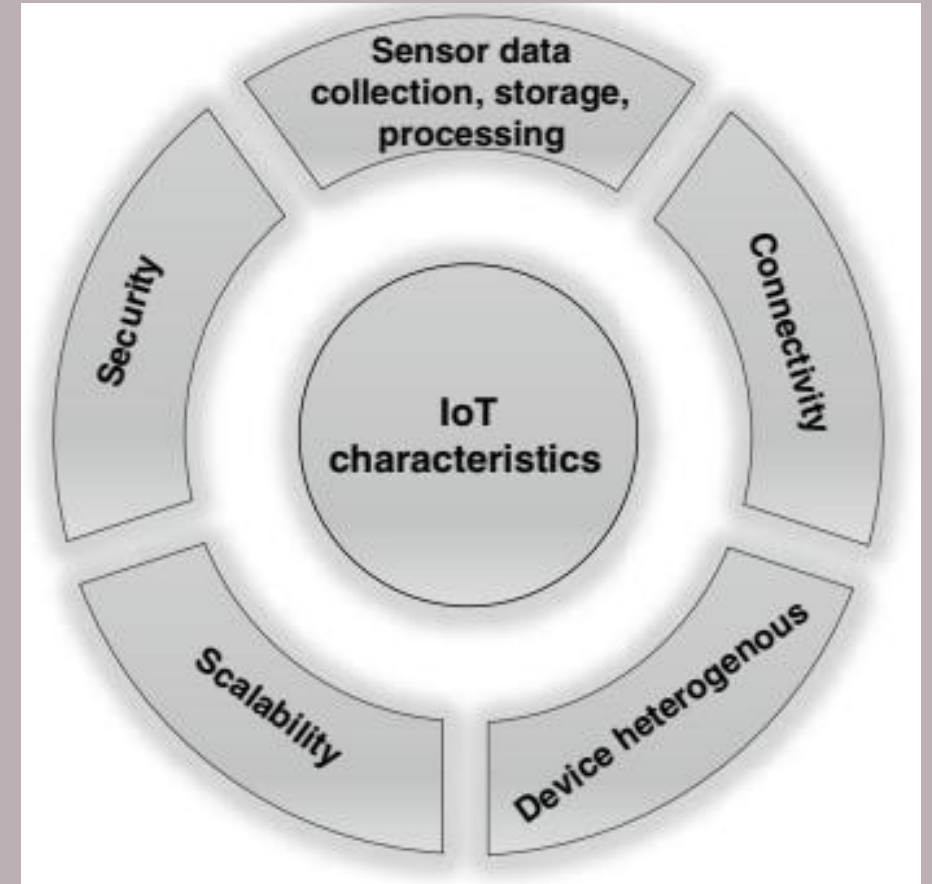
- **Energy-optimized solutions:** For a variety of IoT entities, minimizing the energy to be spent for communication/computing purposes will be a primary constraint. While techniques related to energy harvesting (by means, e.g., of piezo-electric materials or micro solar panels) will relieve devices from the constraints imposed by battery operations, energy will always be a scarce resource to be handled with care. Thereby the need to devise solutions that tend to optimize energy usage (even at the expenses of performance) will become more and more attractive.



Internet of Things

Summary: The key system-level features of the Internet of Things are:

- **Localization and tracking capabilities:** As entities in IoT can be identified and are provided with short-range wireless communications capabilities, it becomes possible to track the location (and the movement) of smart objects in the physical realm. This is particularly important for application in logistics and product life-cycle management, which are already extensively adopting RFID technologies.



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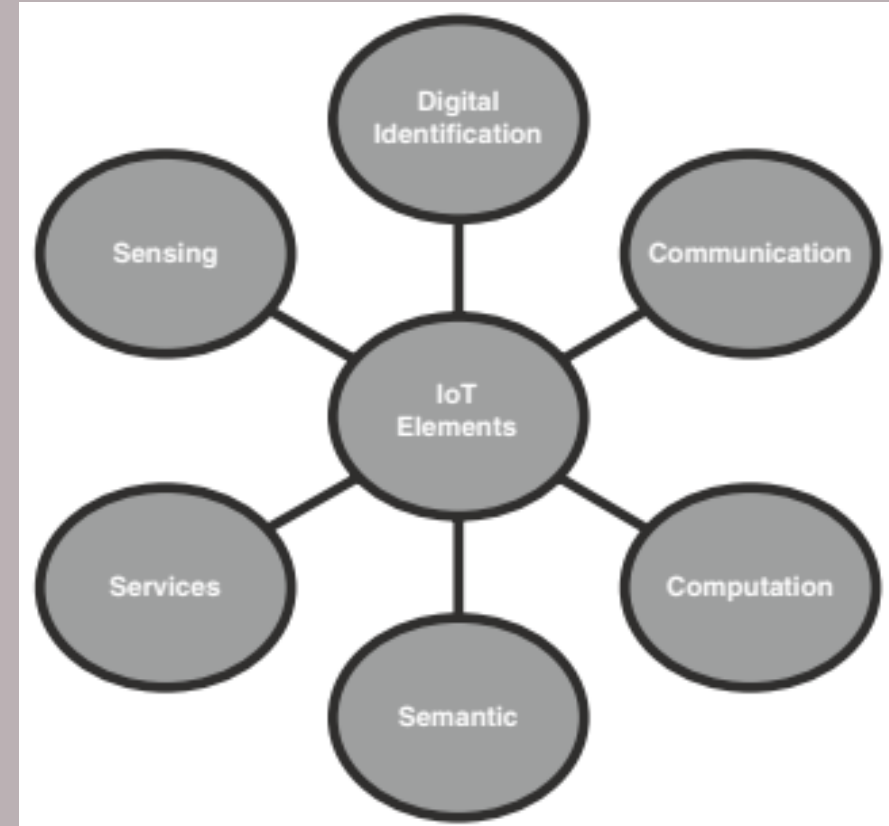
In summary, the key system-level features of the Internet of Things are:

IoT characteristic	Description
Sensor Data Acquisition, Storage, Filtering and Analysis	The plethora of distributed Sensors (or smart things) gather observation of physical environment/entity and direct to Cloud for storage and analytics with an ultimate objective to improve business workflow
Connectivity	IoT has made possible the interconnectivity of Physical and Virtual things with the help of the Internet and global communication infrastructure (that is built using wired and wireless technologies)
Device Heterogeneity and Intelligence	The interoperability of devices (based on different hardware and network platforms) with the provisioning of ambient intelligence at the hardware/software level supports intelligent interactions
Scalability	The plethora of IoT devices connectivity shifts human interactions to device interactions
Security	The security paradigm is required to be implemented at the network level as well as the end-devices level to ensure the security of data

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The IoT paradigm is a multilayer technology that supports meaningful communication of billions of smart things equipped with a processor, sensor/actuator, and communicator.

Considering basic IoT elements (as shown in the figure), the IoT essentially connects a diverse range of hardware devices to a plethora of application domains. The heterogeneity of application and hardware domains imposes varied significant challenges that are essential to meet for the successful deployment of simple and complex IoT systems. In addition to heterogeneity, considering all time ubiquitous connectivity, IoT needs to address a diverse range of issues including scalability, interoperability, security/privacy, and QoS for high traffic/storage needs that ultimately affect the architecture of IoT systems.



IoT Elements

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Enabling Disciplines of IoT

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Enabling Disciplines of IoT

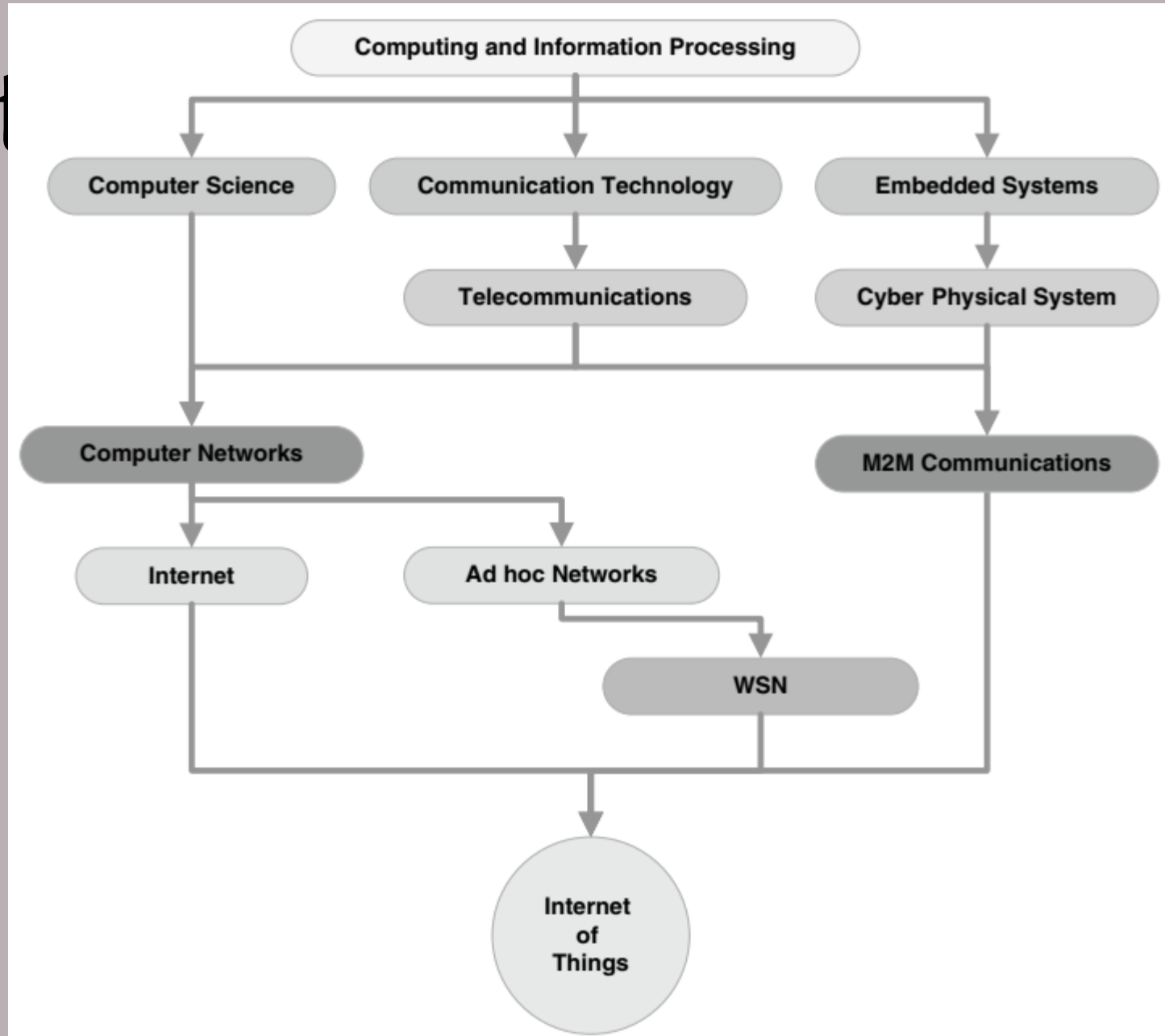
The IoT vision demands significant advances in different fields of ICT (i.e. digital identification technology, communication technology, networking technology, computing technology, and distribution system technology), which are in fact the enabling technologies or fundamental elements of IoT.

IoT is essentially the outcome of various existing technologies used for the collection, processing, inferring, and transmission of data.

Int

Enabling Disciplines of IoT

IoT relationship with peer
technologies



Internet of Things

IoT Enabling Technologies

In general, miniaturization, portability, ubiquitous connectivity, integration of a diverse range of emerging devices, and pervasive availability of digital ecosystems (i.e. Cloud) are the general enablers that play a significant role for enabling IoT systems .

Internet of Things

IoT Enabling Technologies

IoT is enabled by several technologies including

- wireless sensor networks,
- cloud computing,
- big data analytics,
- embedded systems,
- security protocols and architectures,
- communication protocols,
- web services,
- mobile Internet, and
- semantic search engines.

Internet of Things (IoT)

What are the factors making it possible to interconnect more and smaller devices cheaply and easily?

Internet of Things

IoT Enabling Technologies

In general, miniaturization, portability, ubiquitous connectivity, integration of a diverse range of emerging devices, and pervasive availability of digital ecosystems (i.e. Cloud) are the general enablers that play a significant role for enabling IoT systems .

Internet of Things (IoT)

The confluence of several technology and market trends is making it possible to interconnect more and smaller devices cheaply and easily.

1) UBIQUITOUS CONNECTIVITY

Low-cost, high-speed, pervasive network connectivity, especially through licensed and unlicensed wireless services and technology, makes almost everything “connectable”

Internet of Things (IoT)

The confluence of several technology and market trends is making it possible to interconnect more and smaller devices cheaply and easily.

2) WIDESPREAD ADOPTION OF IP-BASED NETWORKING

IP has become the dominant global standard for networking, providing a well-defined and widely implemented platform of software and tools that can be incorporated into a broad range of devices easily and inexpensively.

Internet of Things (IoT)

The confluence of several technology and market trends is making it possible to interconnect more and smaller devices cheaply and easily.

3) COMPUTING ECONOMICS

Driven by industry investment in research, development, and manufacturing, Moore's law continues to deliver greater computing power at lower price points and lower power consumption.

Internet of Things (IoT)

The confluence of several technology and market trends is making it possible to interconnect more and smaller devices cheaply and easily.

4) MINIATURIZATION (NANOTECHNOLOGY)

Manufacturing advances allow cutting-edge computing and communications technology to be incorporated into very small objects. Coupled with greater computing economics, this has fueled the advancement of small and inexpensive sensor devices, which drive many IoT applications.

Internet of Things (IoT)

The confluence of several technology and market trends is making it possible to interconnect more and smaller devices cheaply and easily.

5) ADVANCES IN DATA ANALYTICS

New algorithms and rapid increases in computing power, data storage, and cloud services enable the aggregation, correlation, and analysis of vast quantities of data; these large and dynamic datasets provide new opportunities for extracting information and knowledge.

Internet of Things (IoT)

The confluence of several technology and market trends is making it possible to interconnect more and smaller devices cheaply and easily.

6) RISE OF CLOUD COMPUTING

Cloud computing, which leverages remote, networked computing resources to process, manage, and store data, allows small and distributed devices to interact with powerful back-end analytic and control capabilities.

Internet of Things

Characteristics

Internet of Things

Characteristics

- Efficient, scalable, and associated architecture
- Unambiguous naming and addressing
- Abundance of sleeping nodes, mobile and non-IP devices
- Intermittent connectivity

Internet of Things

Characteristics

- Efficient, scalable, and associated architecture
 - a) The IoT has to be efficient. It has to serve efficiently the requirements of the applications for which they are deployed.
 - b) The IoT has to be scalable in order to accommodate the increasingly large number of things that are being connected to it. Even if the number of sensors and the sensing devices or things are going to increase, the overall network performance of the IoT network should not be compromised.

Internet of Things

Characteristics

- Unambiguous Naming and Addressing
 - There has to be unambiguous naming and addressing architecture. This is very important because we are going to run into a bigger problem with naming and addressing due to the very large number of connected devices. We need a new mechanism for naming and addressing of the different nodes, these physical nodes, the physical objects that are fitted with embedded systems.

Internet of Things

Characteristics

- Abundance of sleeping nodes, mobile and non-IP devices
 - In terms of the resource requirements, the nodes in IoT are typically very low power. They have very low resources and, whenever they are not required, they have to be put to the sleeping mode, they have to go through a sleep cycle.
 - Accordingly, whenever they are not being used, they are not being active. They have to be put in a sleep state and whenever it is required, they have to be made active.

Internet of Things

Characteristics

- Abundance of sleeping nodes, mobile and non-IP devices
 - These devices can be mobile, they can move. So, mobility like this becomes a very important problem in the context of IoT networks. Mobility of the devices and the mobility of the sub networks also is possible. *[For example, a smartwatch you know whoever is wearing the smartwatch, when they move, this node also moves, the smartwatch also moves.]*

Internet of Things

Characteristics

- Abundance of sleeping nodes, mobile and non-IP devices
 - In other words, part of the network becomes mobile and in extreme cases even larger network can also become mobile. Because of this mobility, the IoT device or subnetwork may experience intermittent connectivity in which the device or subnetwork is in connectivity with another device at one instant but is disconnected at a later instant of time.

Internet of Things

Characteristics

- Intermittent connectivity
 - Intermittent connectivity is another problem that has to be taken care of. Opportunistic mobile networks are useful in order to address the problems of intermittent connectivity between the different nodes in the network.

Challenges

Internet of Things

Challenges

The Internet of Things (IoT) has drastically changed the way people live, communicate and do business. There are many benefits of using IoT and many areas where it can be deployed. However, IoT has its fair share of challenges like any other IT and Networking technologies and are listed below:

- 1) *Interoperability*: The term interoperability means the ability of different information systems, devices and applications to access, exchange, integrate and cooperatively use data in a coordinated manner. Interoperability is significantly more challenging for the IoT as it is not (only) about connecting people with people, but about a seamless interaction between devices and people with devices. These devices can differ regarding their technological capabilities.

Internet of Things

Challenges

- 2) *Security and Privacy*: IoT makes everything and everyone locatable and addressable which raises the question of security and privacy. Data integrity, unique identification, and encryption are considered core challenges for IoT, as much of the data being acquired and communicated contain personal information. From the perspective of consumer, IoT device security may open multiple potential risks by enabling unauthorized access, misuse of customer personal information, facilitating attacks on the other integrated systems which may cause individual physical harms.

Internet of Things

Challenges

- 3) *Addressing and Identification:* Uniquely identifying objects is a critical issue for the operation and success of IoT applications. Since millions of smart things will be connected to the Internet, they each will have to be identified through a unique address, on the basis of which they communicate with each other. For this, we need a large addressing space for accommodating all the smart devices, and a unique address for each smart object.

Internet of Everything Internet of Things

3) Addressing and Identification:

- Born in the last century, the internet uses a coding system currently based on 32 bit addresses, which permits the definition of some 4 billion or so separate items.
- Today, electronic product codes for individual items have emerged for use on tiny tags fixed with radio transmitters. Japan's Ubiquitous ID Centre, for instance, has implemented a 128-bit addressing system for tagging individual objects. By some calculations, the Ucode system would allow for a theoretical

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codes to be assigned. This will permit a trillion tags to be assigned every day for a trillion years, and still have some left over

Internet of Things

Challenges

- 4) *Energy Consumption*: Most of the IoT devices are battery powered or use energy harvesting techniques (e.g. solar energy). Communication consumes a lot of energy so they cannot communicate all the time. Hence, designing energy efficient network architecture and intelligent routing mechanism is still a great challenge in IoT networks.

Internet of Things

Challenges

- 5) *Scalability*: Billions of devices get connected to a huge network and large volumes of data are needed to be processed. And the number of connected devices are ever growing, so more devices means the system that stores, analyzes the data from these IoT devices needs to be highly scalable. The raw data obtained from various devices needs big data analytics and cloud storage for interpretation of useful data.

Internet of Things

Challenges

- 6) *Transmission Medium (Communication Protocols)*: IoT networks use different types of technology to transmit or receive the data such as RFID, Bluetooth, Zigbee, LoraWAN, Sigfox, etc. The traditional problems associated with transmission media (e.g., high error rate, bandwidth, fading, inference, etc.) exists for IoT as well. Each transmission medium requires specialized energy, network hardware, the bandwidth that has to be compatible with that medium. Therefore, optimizing the TM is a challenge in IoT applications to sustain and prolong the lifetime of networks.

Internet of Things

Challenges

- 7) *Big Data and data analytics*: IoT and its large number of sensors is going to trigger a deluge of data that must be processed. This data will provide critical information and insights if it can be processed in an efficient manner. The challenge, therefore, is evaluating these massive amounts of data arriving from different sources in various forms and doing so in a timely manner.